Comparing Three GPS Noise Filters: Which Best Recovers a Clean Signal Buried in Noise

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Abstract

Every Global Positioning System (GPS) measurement carries noise, and on a real station a filter cannot be evaluated: the true path is unknown. A clean signal was therefore built in R (600 samples plus noise of a known size) and each filter scored by root mean square error (RMSE). With noise of standard deviation (sd) 5 millimetres (mm) the Kolmogorov-Zurbenko (KZ) filter gave an RMSE of 0.890 mm, below a Butterworth low-pass (1.671 mm) and a Simple Moving Average (SMA, 2.239 mm), against 4.913 mm unfiltered, and stayed lowest at all four noise levels. On 7225 daily solutions from the real station with code NYCI (Long Island, New York) no truth exists, so roughness replaced RMSE: 0.166 mm after KZ, 6.525 mm raw. RMSE and roughness answer different questions: a smooth line can still sit in the wrong place. Settings were fixed, not tuned. For this signal, KZ is the filter this study would apply to a real GPS trajectory.

Introduction and Questions

- A continuous GPS station reports East, North and Up once a day, and the line never sits still (Fig. 1).
- Noise comes from the atmosphere, from signals bouncing off surfaces, and from the receiver.
- A rocket or a vehicle navigating on a noisy position can follow a miscalculated path.
- On real data no filter can be evaluated, so a clean signal was built, known noise added, and each filter's return to it measured.
- Questions. Which filter recovers the clean signal with the lowest RMSE? Does it stay lowest as noise grows? What happens on a real station?

Methodology

- Clean signal. 600 samples, clean = 0.03 t + 8 sin(2 pi t / 150), in mm, where t is the sample number.
- Added noise. Gaussian, mean 0, sd 5 mm, seed fixed at 42.
- Filtering. SMA window 15; KZ window 15, three passes; Butterworth order 3, cutoff 0.05.
- Scoring. RMSE from the known truth, samples 15 to 600.
- Noise sweep. Repeat the whole test at noise sd 2, 5, 10 and 20 mm.
- Real data. Same settings, Up of GPS station NYCI, 7225 daily solutions, 25 Apr 2006 to 4 Jul 2026.
- Change of metric. No truth on a real record, so roughness (sd of the day-to-day change) replaces RMSE.

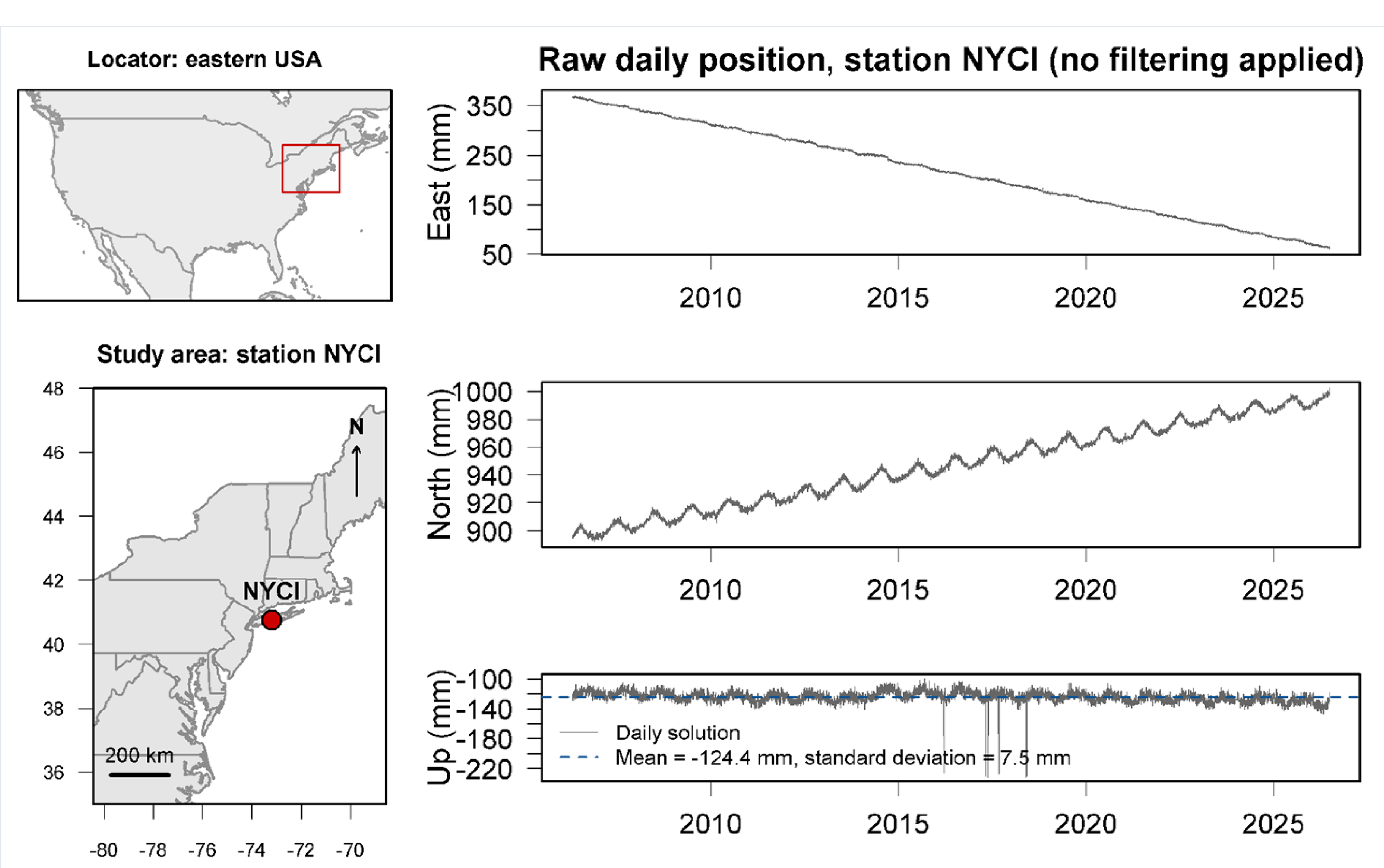

Fig. 1. Maps of the study area and line plots of the raw daily East, North and Up positions in millimetres (vertical axis) against date (horizontal axis) at GPS station NYCI, Long Island, New York. One solution is one daily position estimate, and the scatter in these 7225 solutions is the noise the filters must remove.

Results

- The noise was added on purpose, so the RMSE in Table 1 is a real recovery error.
- At noise sd 5 mm, KZ landed 0.890 mm from the truth and removed 81.9 percent of the error.
- Butterworth 1.671 mm (66.0 percent), SMA 2.239 mm (54.4 percent), no filter 4.913 mm (Fig. 2).
- KZ stayed closest at every level: 0.479, 0.890, 1.654 and 3.224 mm at sd 2, 5, 10 and 20 mm (Fig. 3).
- A single lowest RMSE could be chance; the lowest RMSE at all four noise levels is a consistent result.
- On real station NYCI there is no known truth, so the score changes to roughness (Fig. 4).
- Roughness (mm) fell from 6.525 raw to 0.166 after KZ, against 0.422 Butterworth and 0.550 SMA.

Table 1. RMSE of each filter from the known clean signal and the percent of error removed: 600 samples, noise sd 5 mm, window 15, scored on samples 15 to 600.

Series	RMSE vs truth (mm)	Error removed
No filter (raw noisy)	4.913	0.0 %
Simple Moving Average	2.239	54.4 %
Butterworth low-pass	1.671	66.0 %
Kolmogorov-Zurbenko	**0.890**	**81.9 %**

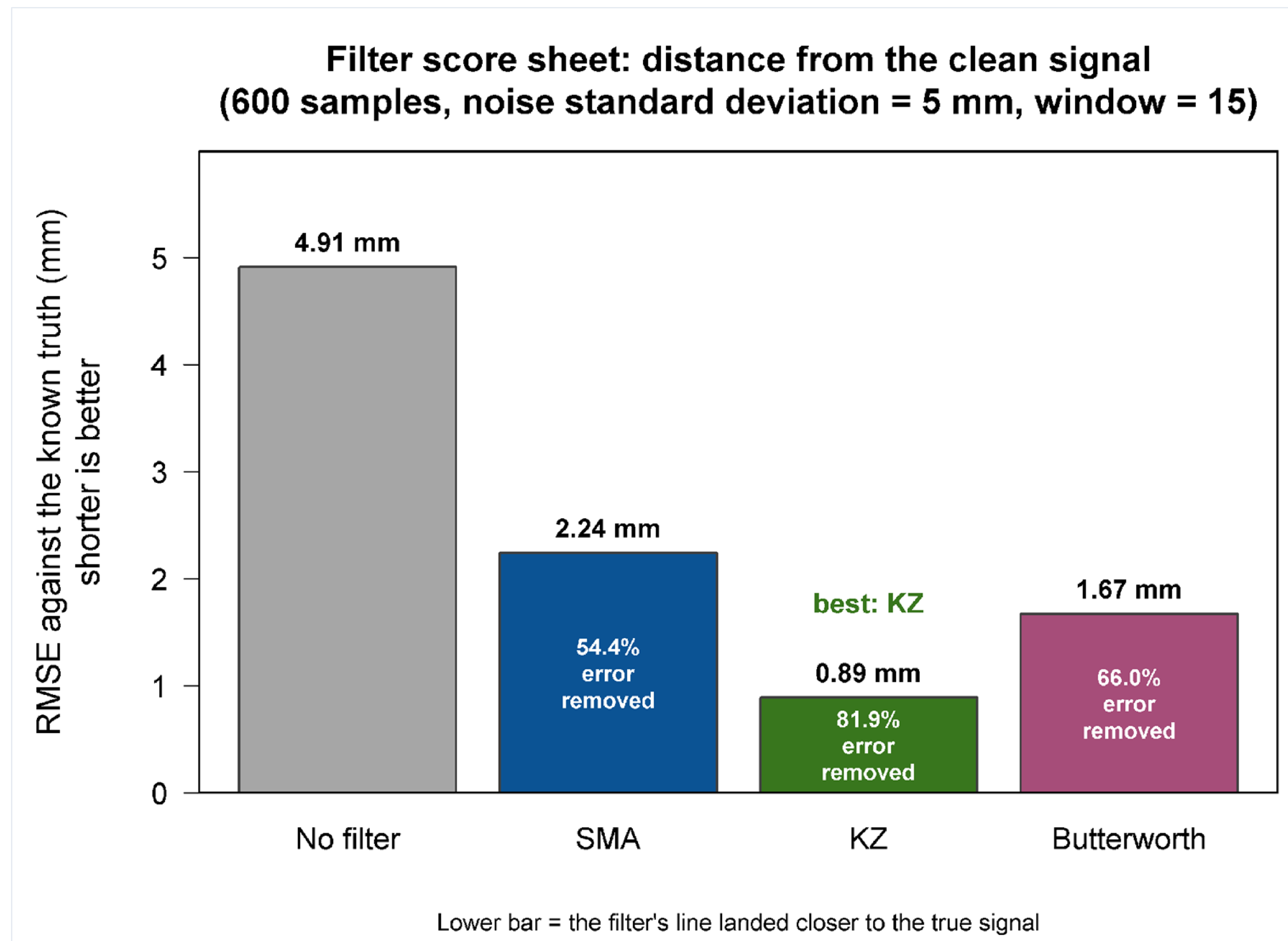

Fig. 2. Bar chart of each filter's RMSE from the known clean signal at noise sd 5 mm, one bar per series. The KZ bar is the shortest at 0.890 mm, against 4.913 mm with no filter.

Discussion and Conclusions

- KZ gave the lowest RMSE in the controlled test and at every noise level, so the choice is measured, not assumed.
- RMSE and roughness answer different questions and must not be mixed.
- RMSE says how close a filter landed to the truth, and only a known truth makes that possible.
- Roughness says only how smooth a line is. A very smooth line can still sit in the wrong place.
- For this signal, this noise and these settings, KZ is the filter this study would apply to a real GPS trajectory.

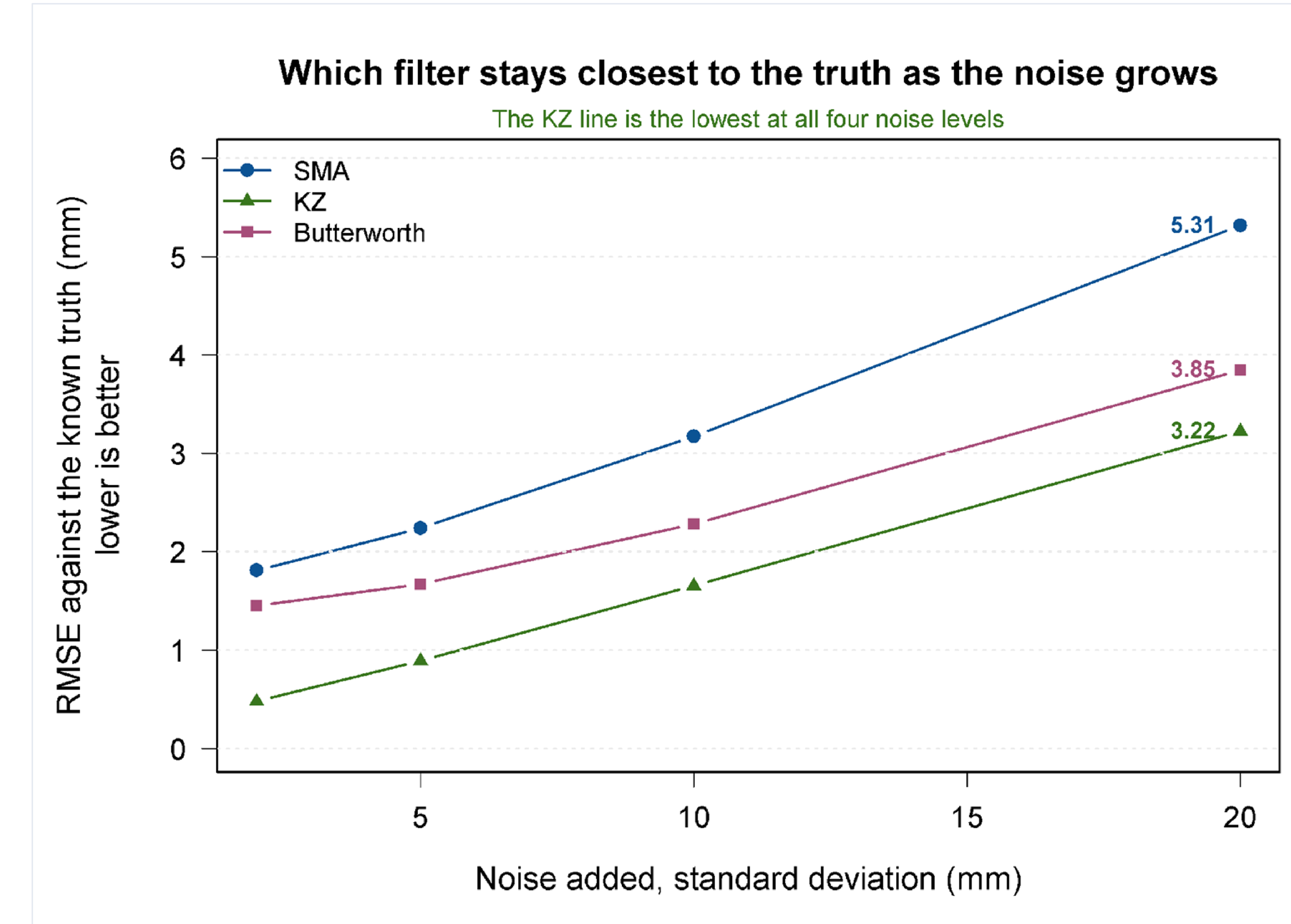

Fig. 3. Line plot of each filter's RMSE from the known clean signal as the added noise grows from sd 2 mm to 20 mm. The KZ line stays lowest at all four levels.

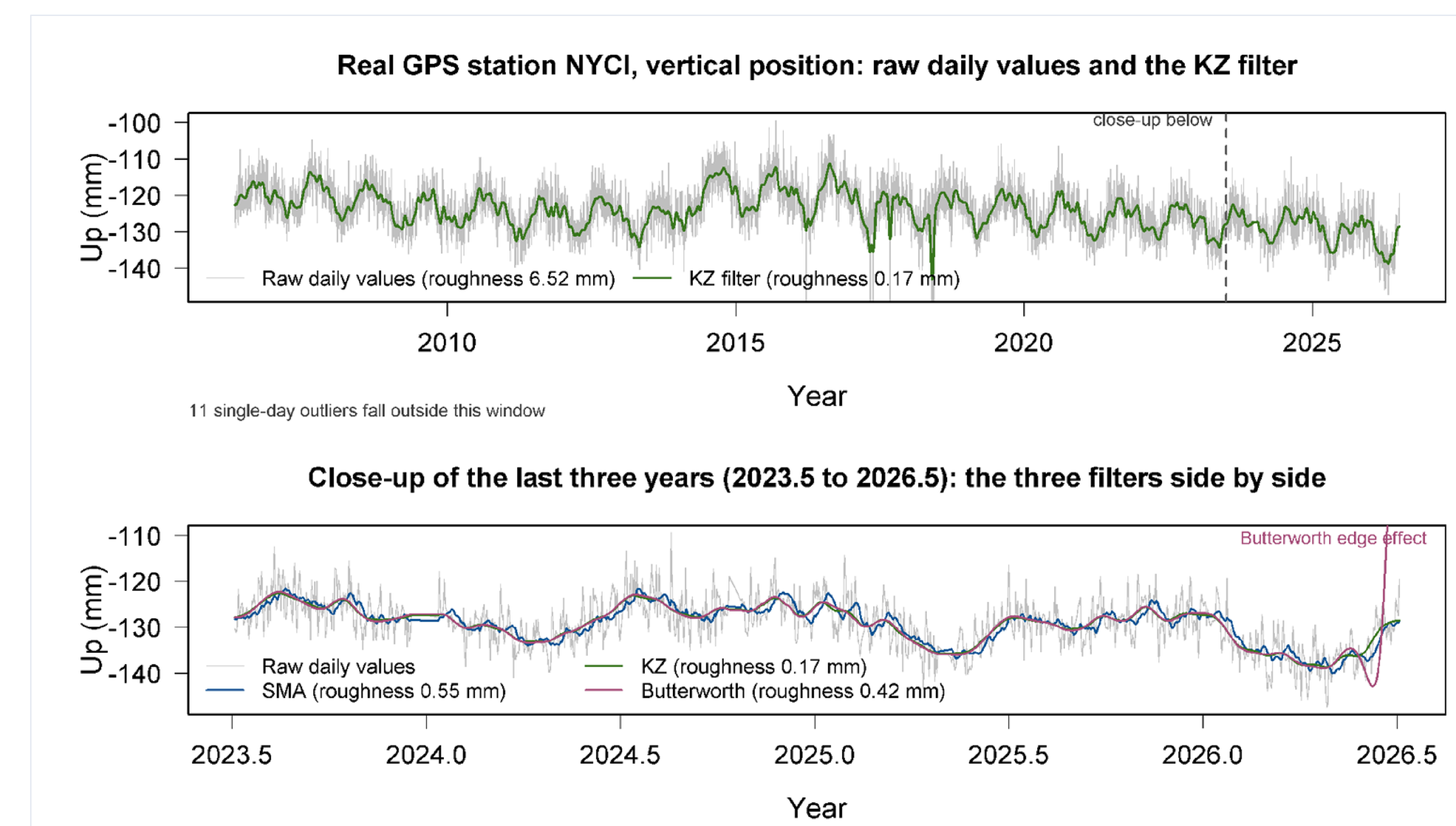

Fig. 4. Line plots of the real NYCI vertical position (vertical axis, millimetres) against date (horizontal axis), with the three filtered lines on top: the whole 2006 to 2026 record, then a close-up of the last three years. Roughness falls from 6.525 mm raw to 0.166 mm after KZ.

LIMITATIONS

Fixed settings (window 15, three passes, cutoff 0.05), never tuned, so another window might beat KZ.
The Butterworth line jumps at the end of the real record, where no neighbouring values are left.
Eleven single days in NYCI lie more than five sd from the mean and were left in place.
Roughness measures smoothness, not accuracy, and NYCI is a fixed ground station, not a rocket, so the step to a launch trajectory rests on published work.

SIGNIFICANCE

Cleaner GPS trajectories mean better estimates of where a rocket, a vehicle or a piece of ground truly went. Selecting the filter by measurement, rather than by assumption, is what makes the reported accuracy verifiable.

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KEYWORDS AND CONTACT

GPS time series, noise suppression, Kolmogorov-Zurbenko filter, Butterworth low-pass, moving average, RMSE
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